

MAT 310, Handout 5: Law of Total Probability, Bayes' Rule, Random Sampling (2.9, 2.10, 2.12)

Calculating Probability of an Event: Event Composition Method

1. Write A as a composition of other events (e.g., using unions, intersections, complements)
2. Calculate $P(A)$ using Probability Laws

Example 1. A lie detector shows a positive reading (indicating a lie) 10% of the time when a person is telling the truth and 90% of the time when a person is lying. Suppose two people are suspects in a one-person crime and one is guilty and will lie. Assuming the lie detector operates independently, find the probability that the detector:

$A = \{\text{shows a positive reading for both suspects}\}.$

$B = \{\text{is positive for the guilty suspect and negative for the innocent suspect}\}.$

$C = \{\text{is positive for the innocent suspect and negative for the guilty suspect}\}.$

$D = \{\text{gives a positive reading for either or both of the two suspects}\}.$

Definition: Law of Total Probability Suppose $B_1, B_2, \dots, B_k$ form a partition of $\mathbb{S}$, that is,

1. $B_1 \cup B_2 \cup \dots \cup B_k = \mathbb{S}$ &

2. $B_i \cap B_j = \emptyset$ for $i \neq j$.

Then, if $P(B_i) > 0$ for all i ,

$$P(A) =$$

Proof:

Example 2. Show $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$.

Definition: Bayes' Rule Suppose $B_1, B_2, \dots, B_k$ is a partition of $\mathbb{S}$ such that $P(B_i) > 0$ for $i = 1, \dots, k$. Then,

$$P(B_j|A) =$$

Proof:

Example 3. According to a CDC study, about 0.2% of babies in the US are born with hearing loss. Newborns typically complete a hearing screening before leaving the hospital. These screenings have false positive rates around 5% and false negative rates around 0.5%. What is the probability that a baby with a positive test actually has hearing loss?

Example 4. There are 3 boxes, each with 2 compartments. The first box has a bar of gold in both compartments, the second box has a bar of silver in both compartments, and the third box has a bar of gold in one compartment and a bar of silver in the other. One box is chosen at random, then one of its compartments is opened at random. If the bar inside is gold, what is the probability that the bar in the other compartment is also gold?

Example 5. Suppose 20% of cabs in a city are blue and the other 80% are green. An eyewitness of a night-time hit and run accident claims that the cab involved was blue. However, under night-time conditions, only 75% of people can tell the difference between blue and green. Based on this information, what is the probability that the cab at fault was blue?

Example 6. There are 3 coins in a box: a two-headed coin, a fair coin, and a weighted coin that shows heads $2/3$ of the time. You randomly select the coin, then flip the coin once. The result is a head. What is the probability you selected the two-headed coin.

Another Approach (Tree Diagram):

A **random sample** of size n from a population of size N is one in which each of the $\binom{N}{n}$ subsets of size n has an equal probability of being selected as the sample.

A **pseudo-random number generator** is mathematically designed to generate numbers that are statistically random. These number sequences are seeded and have an extremely large (albeit finite) cycle.